

SERGIO TASCON MORALES

PhD Student



PROFILE

A goal-oriented third-year Ph.D. student at the University of Bern. Passionate about Computer Vision, Natural Language Processing and their combination. With a structured and organized approach and the ability to work under pressure, I deliver results on time. As a natural team player, I enjoy and value different perspectives, personalities and cultural backgrounds.



WORK EXPERIENCE

02.2020 - 10.2020
Heidelberg, DE

Research Intern
Mediri GmbH

- > Developing my master thesis on segmentation of multiple sclerosis lesions using a combination of CNNs and RNNs.
- > Leveraging temporal information provided by longitudinal MRI data to improve segmentation quality.

03.2016 - 07.2016
Maulburg, DE

Research Intern
Endress & Hauser

- > Evaluating Python as an alternative for future use in the company.
- > Analyzing models for pressure sensor linearization.



EDUCATION

11.2020 - present
Bern, CH

PhD Student
University of Bern

- > Developing solutions in the topic of Visual Question Answering (VQA).

09.2018 - 09.2020
Le Creusot, FR
Cassino, IT
Girona, ES
Heidelberg, DE

Master in Medical Imaging and Applications (MAIA)
University of Burgundy, University of Cassino, University of Girona

- > Major academic projects include:
 - Left ventricle segmentation using K-Means and Active Contours
 - Classification of Chest X-Ray images using Deep Learning
 - Segmentation and classification of breast mass lesions using grayscale morphological operations and decision trees
 - Classification of skin lesions using classical machine learning and deep learning
 - 3D Harris corner detector

10.2015 - 03.2016
Ilmenau, DE

Exchange Semester
Ilmenau University of Technology

- > Exchange semester during Young Engineers program of the German Academic Exchange Service (DAAD).

02.2011 - 12.2017
Cali, CO

Electronic Engineering
Universidad del Valle

- > Main focus on machine learning and computer vision.

INFO



E-mail

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Website

sergiotasconmorales.github.io



Nationality

Colombian



Marital status

Single

COMPUTER SKILLS

Python

PyTorch

C/C++

Matlab

Linux

Git

LANGUAGES

Spanish	Native
English	Advanced (C1)
German	Advanced (C1)
Italian	Intermediate (B1)
Turkish	Basic (A1)
Russian	Basic (A1)



PUBLICATIONS

- 2023** **Localized Questions in Medical Visual Question Answering** [🔗](#)
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: P. Márquez-Neila, R. Sznitman. [Accepted]
- 2023** **Logical Implications for Visual Question Answering Consistency** [🔗](#)
Conference on Computer Vision and Pattern Recognition (CVPR). Co-authors: P. Márquez-Neila, R. Sznitman.
- 2022** **Consistency-Preserving Visual Question Answering in Medical Imaging** [🔗](#)
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: P. Márquez-Neila, R. Sznitman.
- 2020** **Multiple Sclerosis Lesion Segmentation Using Longitudinal Normalization and Convolutional Recurrent Neural Networks** [🔗](#)
Workshop on Machine Learning In Clinical Neuroimaging at the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: S. Hoffman, M. Treiber, D. Mensing, A. Oliver, M. Günther, J. Gregori.



HONORS & AWARDS

- 2020**
Le Creusot, FR **Best master thesis and best student awards**
University of Burgundy, University of Cassino, University of Girona
- 2018**
Girona, ES **Erasmus+ grant**
European Union
- 2017**
Cali, CO **Bachelor final project with distinction**
Universidad del Valle
- 2015**
Cali, CO **Scholarship Young Engineers**
German Academic Exchange Service (DAAD)
- 2009**
Ginebra, CO **Best high school graduate**
Institución Educativa Inmaculada Concepción



HOBBIES & INTERESTS



Violin



Hiking



Drawing



Reading



Football